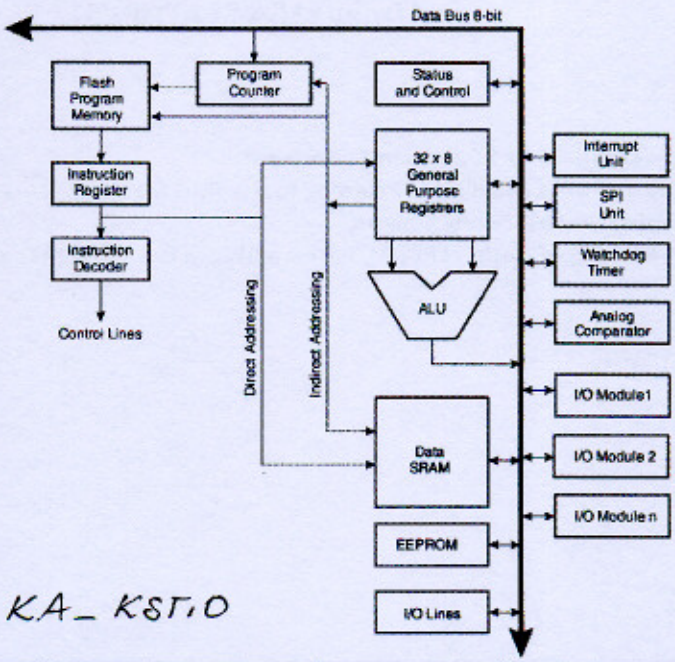
STRUKTURY WBUDOWANE

1

oprac. K. Arnold
KSTiO

Block Diagram of the AVR MCU Architecture



KA_KSTiO

ATmega16A FEATURES

- High-performance, Low-power AVR® 8-bit Microcontroller
- Advanced RISC Architecture
  - –131 Powerful Instructions – Most Single-clock Cycle Execution
  - –32 x 8 General Purpose Working Registers
  - –Fully Static Operation
  - –Up to 16 MIPS Throughput at 16 MHz
  - –On-chip 2-cycle Multiplier
- High Endurance Non-volatile Memory segments
  - –16K Bytes of In-System Self-programmable Flash program memory
  - –512 Bytes EEPROM
  - –1K Byte Internal SRAM
  - –Write/Erase Cycles: 10,000 Flash/100,000 EEPROM
  - –Data retention: 20 years at 85°C/100 years at 25°C(1)
  - –Optional Boot Code Section with Independent Lock Bits
    - •In-System Programming by On-chip Boot Program
    - •True Read-While-Write Operation
  - –Programming Lock for Software Security

KA_KST.0

3

ATmega16A FEATURES

- JTAG (IEEE std. 1149.1 Compliant) Interface
  - –Boundary-scan Capabilities According to the JTAG Standard
  - –Extensive On-chip Debug Support
  - –Programming of Flash, EEPROM, Fuses, and Lock Bits through the JTAG Interface

KA_KST.0

4

ATmega16A FEATURES

- Peripheral Features
  - –Two 8-bit Timer/Counters with Separate Prescalers and Compare Modes
  - –One 16-bit Timer/Counter with Separate Prescaler, Compare Mode, and Capture Mode
  - –Real Time Counter with Separate Oscillator
  - –Four PWM Channels
  - –8-channel, 10-bit ADC
- 8 Single-ended Channels
- 7 Differential Channels in TQFP Package Only
- 2 Differential Channels with Programmable Gain at 1x, 10x, or 200x
  - –Byte-oriented Two-wire Serial Interface
  - –Programmable Serial USART
  - –Master/Slave SPI Serial Interface
  - –Programmable Watchdog Timer with Separate On-chip Oscillator
  - –On-chip Analog Comparator

KA_ KST. O

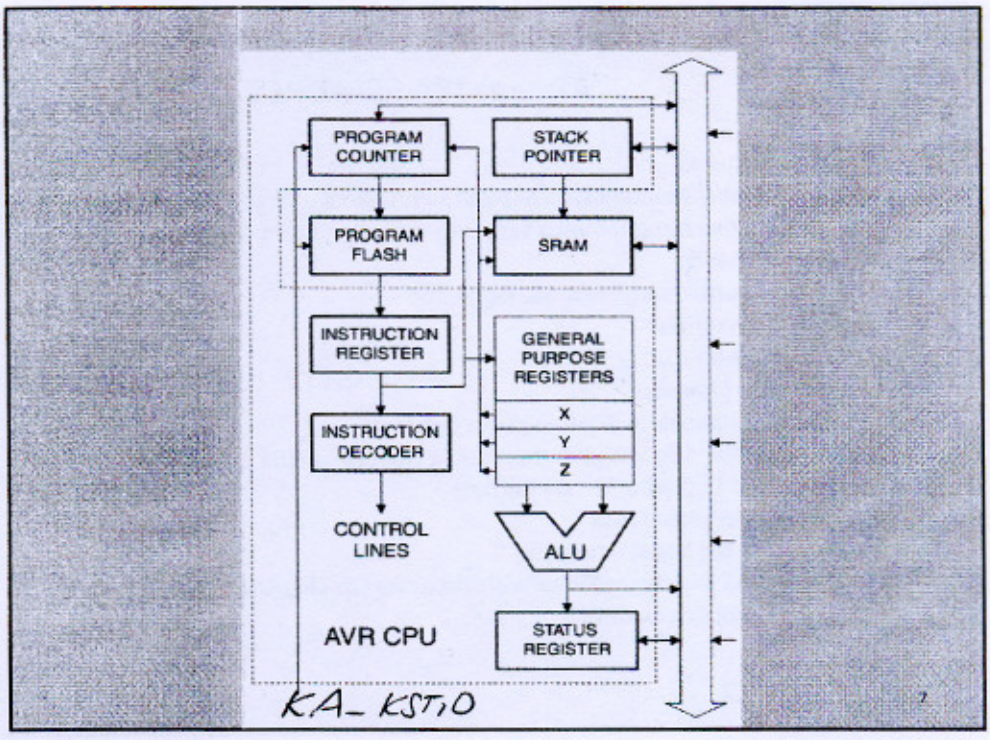
5

ATmega16A FEATURES

- Special Microcontroller Features
  - –Power-on Reset and Programmable Brown-out Detection
  - –Internal Calibrated RC Oscillator
  - –External and Internal Interrupt Sources
  - –Six Sleep Modes: Idle, ADC Noise Reduction, Power-save, Power-down, Standby and Extended Standby
- I/O and Packages
  - –32 Programmable I/O Lines
  - –40-pin PDIP, 44-lead TQFP, and 44-pad QFN/MLF
- Operating Voltages
  - –2.7 - 5.5V for ATmega16A
- Speed Grades
  - –0 - 16 MHz for ATmega16A
- Power Consumption @ 1 MHz, 3V, and 25°C for ATmega16A
  - –Active: 0.6 mA
  - –Idle Mode: 0.2 mA
  - –Power-down Mode: < 1µA

KA_ KST. O

6



SREG – AVR Status Register

7	6	5	4	3	2	1	0
I	T	H	S	V	N	Z	C
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- Bit 7 – I: Global Interrupt Enable
- Bit 6 – T: Bit Copy Storage
- Bit 5 – H: Half Carry Flag
- Bit 4 – S: Sign Bit, S = N ⊕ V
- Bit 3 – V: Two's Complement Overflow Flag
- Bit 2 – N: Negative Flag
- Bit 1 – Z: Zero Flag
- Bit 0 – C: Carry Flag

KA_KST,0

8

AVR CPU General Purpose Working Registers

7 0	Addr.	
R0	$00	
R1	$01	KA_KST1,0
R2	$02	
...		
R13	$0D	
R14	$0E	
R15	$0F	
R16	$10	
R17	$11	
...		
R26	$1A	X-register Low Byte
R27	$1B	X-register High Byte
R28	$1C	Y-register Low Byte
R29	$1D	Y-register High Byte
R30	$1E	Z-register Low Byte
R31	$1F	Z-register High Byte

SPH and SPL – Stack Pointer High and Low Register

SPH

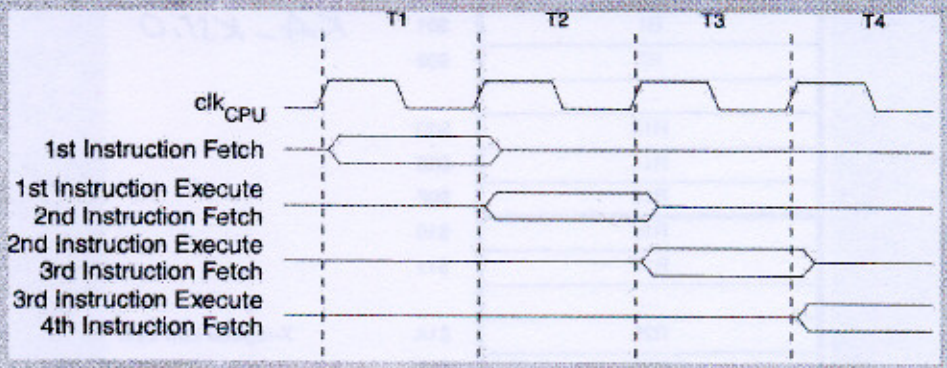
15	14	13	12	11	10	9	8
SP15	SP14	SP13	SP12	SP11	SP10	SP9	SP8

SPL

SP7	SP6	SP5	SP4	SP3	SP2	SP1	SP0
7	6	5	4	3	2	1	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

KA_KST1,0

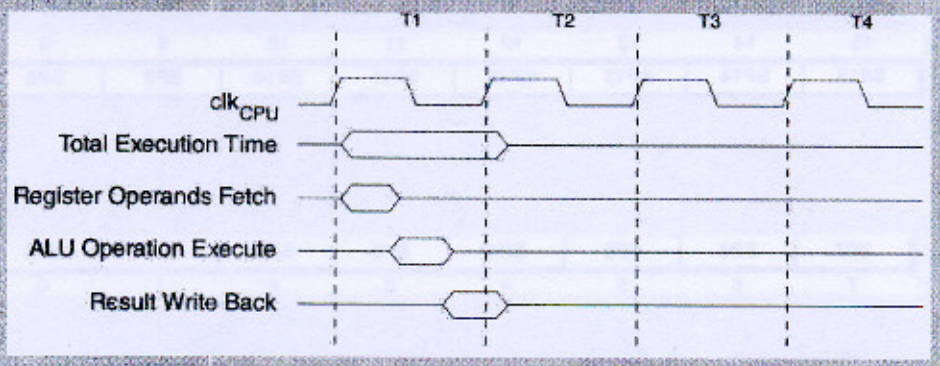
THE PARALLEL INSTRUCTION FETCHES AND INSTRUCTION EXECUTIONS



KA_KST,0

11

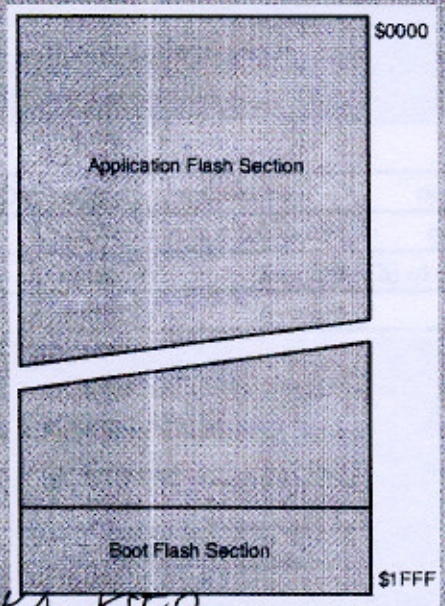
SINGLE CYCLE ALU OPERATION



KA-KST, 0

12

ATmega16A - PROGRAM MEMORY MAP



0x2000
=
0b 0010 0000 0000 0000
=
2^{13}
=
8192
=
8 kstów 16-bitowych
=
16KB

KA_KST,0

19

ATmega16A - PROGRAM MEMORY ADDRESSING

Adres słowa	Adres bajtu	Adres bajtu
0000		
0001	00010	00011
0010	00100	00101
0011	00110	00111
0100	01000	01001

KA_ICST,0

14

AVR – JMP INSTRUCTION SET

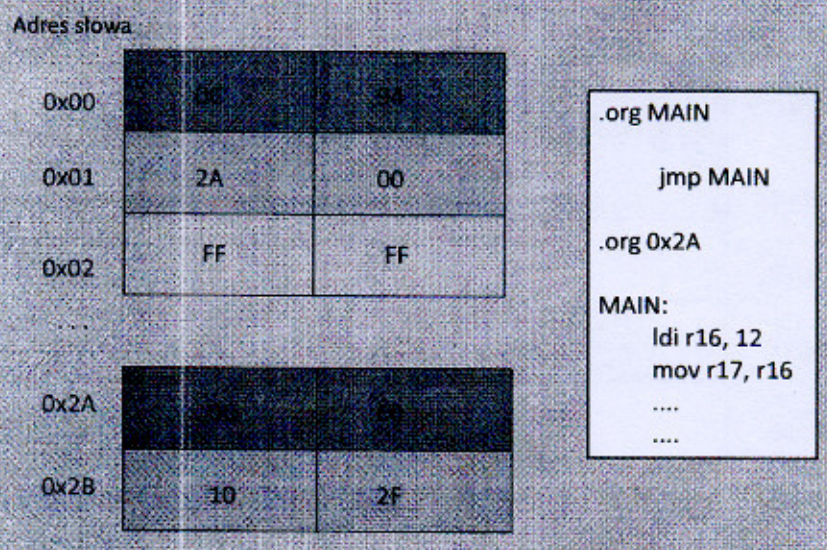
BRANCH INSTRUCTIONS

Mnemonics	Operands	Description	Operation	Flags	#Clocks
RJMP	k	Relative Jump	PC ← PC + k + 1	None	2
IJMP		Indirect Jump to (Z)	PC ← Z	None	2
JMP	k	Direct Jump	PC ← k	None	3

KA_KST0

15

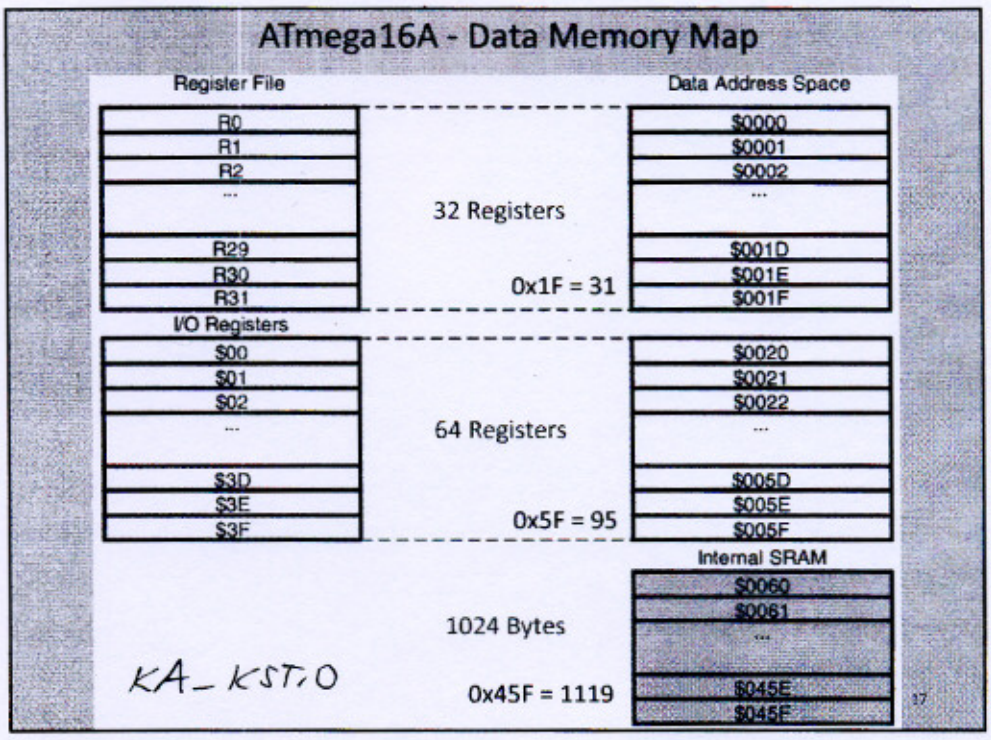
AVR – RESTART EXAMPLE

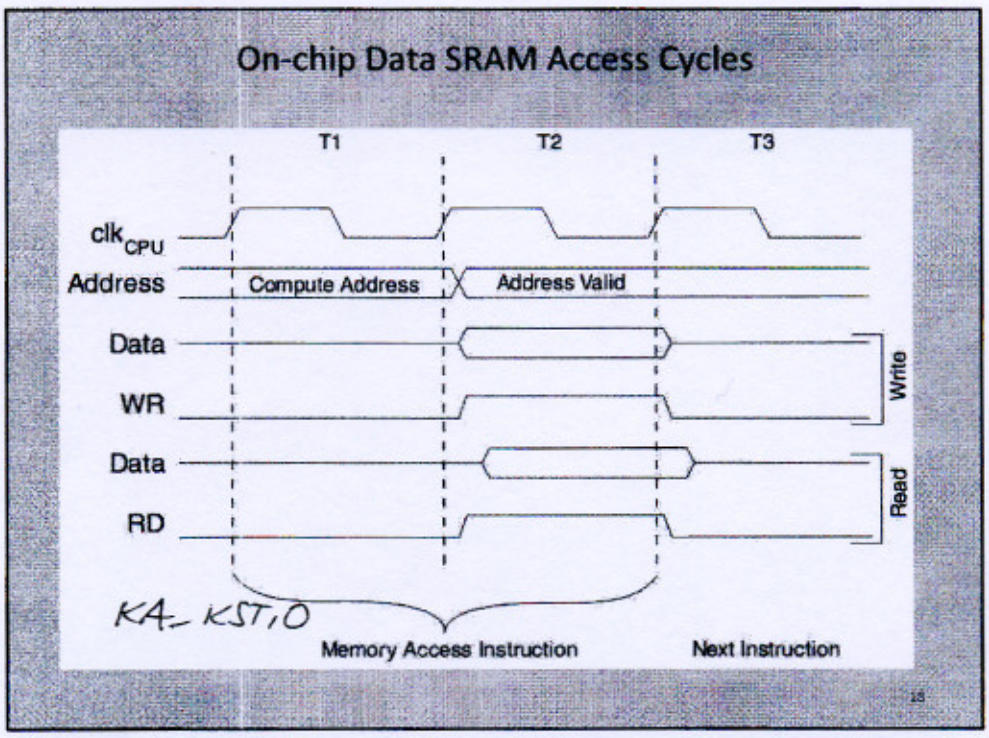



```
.org MAIN
        jmp MAIN
.org 0x2A
MAIN:
        ldi r16, 12
        mov r17, r16
        ....
        ....
```

KA_KST0

16


ATmega16A - Data Memory Map
Register File
Data Address Space
R0
R1
R2
...
R29
R30
R31
32 Registers
0x1F = 31
$0000
$0001
$0002
...
$001D
$001E
$001F
I/O Registers
$00
$01
$02
...
$3D
$3E
$3F
64 Registers
0x5F = 95
$0020
$0021
$0022
...
$005D
$005E
$005F
Internal SRAM
1024 Bytes
0x45F = 1119
$0060
$0061
...
$045E
$045F
KA_KST:0
17


On-chip Data SRAM Access Cycles
T1
T2
T3
clk CPU
Address
Compute Address
Address Valid
Data
WR
Write
Data
RD
Read
KA_KST:0
Memory Access Instruction
Next Instruction
18